What We Actually Measured: A Correctness Audit, a Virtualization Measurement-Bias Case Study, and a 2024 Strassen Variant, Benchmarked

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Draft – September 9, 2026

Abstract

We revisit a published AVX-512 dense-matrix-multiplication (DGEMM) and Strassen implementation with three goals: check that it does what it claims, check that its reported performance wins are real hardware effects rather than measurement artifacts, and extend its numerical- accuracy story with a result from the current literature that had not yet been benchmarked anywhere. First, a correctness audit finds and fixes four real, execution-verified bugs, the most serious a packing error that silently computed $C = B \cdot A$ instead of $C = A \cdot B$ for non-square shapes. Second, a large (≈2×) OpenMP dynamic-scheduling performance win, initially measured on a virtualized development environment, is shown – via a controlled, properly repeated measurement, including in the heterogeneous performance/ efficiency-core case scheduling theory predicts should favor it most – not to reproduce; only raw core placement (performance core vs. efficiency core, a ≈2× effect) survives repetition, extending Mytkowicz et al.'s general measurement-bias methodology into a domain – DGEMM/OpenMP scheduling claims – it has not previously been applied to. Third, we implement Dumas, Pernet & Sedoglavic's 2024 accuracy-optimized alternate-basis Strassen algorithm as a real recursive block algorithm – symbolically verifying its formula against the paper's own straight-line program first, which catches a real subscript-convention transcription error – and measure, for what appears to be the first time, both its accuracy improvement (confirmed: consistently lower Frobenius-relative error than classical Strassen at matched recursion depth) and its performance cost (≈20% fewer GFLOP/s at N=2048, depth 3), a trade-off the original paper does not report. All results are backed by a public, locally-reproducible harness and dataset, organized so each numeric claim in this paper traces back to a named, re-runnable script and CSV.

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1 Introduction

1.1 Motivation

High-performance DGEMM and fast-matrix-multiplication engineering has a deep, well-cited lineage: Goto & van de Geijn's multilevel blocking and packing model [10], generalized into the portable BLIS framework [14], and Huang et al.'s demonstration that Strassen's extra additions can be fused into an already-optimized DGEMM's packing routines at essentially no extra workspace cost [11]. What is comparatively rare in that literature is independent, adversarial re-execution: taking a specific published implementation, running it, and checking whether it actually does what its paper claims. This project started as exactly that exercise – auditing a published AVX-512

DGEMM/Strassen implementation against its own paper – and the audit surfaced real bugs (Section 4), which is itself evidence that this kind of check is worth doing more often, not just for this one implementation.

The audit also surfaced a methodological question with broader implications than this one codebase: a large, apparently clean OpenMP scheduling performance win, measured once on a virtualized development environment, turned out not to survive a properly repeated measurement (Section 5). This is a domain-specific instance of a general phenomenon Mytkowicz et al. documented in 2009 – that "insignificant" experimental-setup choices can flip a benchmark's conclusion [13] – and, as far as our own literature search found (Section 2), it has not previously been demonstrated as a targeted case study for a DGEMM/OpenMP scheduling claim specifically. Given how routinely such benchmarks are now run inside cloud VMs and WSL2-style virtualized dev environments, that gap seemed worth closing with a concrete, repeatable example rather than a general warning.

Finally, in the course of extending this project's own numerical- accuracy work, we found a genuinely new (2024) result in the fast- matrix-multiplication literature – an alternate-basis Strassen variant with a provably better accuracy bound [9] – that, as far as we could determine, had never been implemented as a real recursive block algorithm or benchmarked on hardware under realistic packed/blocked execution; the original paper reports only asymptotic operation counts and an accuracy-vs-size plot, no wall-clock or phase-timed performance data at all. Section 6 closes that gap and reports the accuracy/performance trade-off its own authors did not measure.

1.2 Contributions

1. **A correctness audit of a published AVX-512 DGEMM/Strassen implementation**, finding and fixing four real bugs with execution evidence (not just static reading), the most significant a packing bug that silently computed $B \cdot A$ instead of $A \cdot B$; a second-pass, repo-wide sweep found and fixed six further bugs, including a regression introduced by this audit's own first-pass fix. See Section 4.

2. **A measurement-bias case study**: a large (2×-class) OpenMP dynamic-scheduling performance win does not reproduce under proper repeated measurement – including in the heterogeneous performance/efficiency-core case theory predicts should favor it most; only raw core placement survives repetition. See Section 5.

3. **Dumas et al.'s 2024 alternate-basis Strassen, implemented and measured**: a real recursive block algorithm built on infrastructure identical to a classical-Strassen baseline, symbolically verified before implementation (catching a real subscript-convention transcription bug), then measured for both accuracy (confirms the paper's claim, under conditions the paper itself never tested) and performance cost (a number nobody has published). See Section 6.

4. Supporting material: a from-scratch AVX2 DGEMM tuned against production BLAS, hardware-counter root-cause analysis explaining *why* several classical performance results hold, cross- compiler codegen differences, a from-scratch classical-Strassen rebuild with a measured optimized-vs-optimized crossover, and a rectangular/irregular-shape performance-cliff finding. See Section 7.

5. A public, reproducible harness and dataset backing every number in this paper. See Section 9.

2 Related Work

2.1 High-performance GEMM engineering

Goto & van de Geijn's "Anatomy of High-Performance Matrix Multiplication" [10] establishes the multilevel cache-blocking and operand-packing model (an $M_C \times K_C$ panel of A and a $K_C \times N_C$ panel of B, packed into contiguous buffers and driven through a small register-blocked microkernel) that underlies essentially all modern high-performance DGEMM, including both the implementation we audit and our own from-scratch AVX2 microkernel work (Section 7). Van Zee & van de Geijn's BLIS framework [14] generalizes that model so that only a small, architecture-specific microkernel needs re-tuning per target – a design pattern our own portable, runtime-configurable blocking layer (FINDINGS.md Part B.12) independently converges on at smaller scale. These are the ceiling our own plain-DGEMM measurements are stated relative to, not a target we claim to beat.

2.2 Practical fast matrix multiplication

Huang, Smith, Henry & van de Geijn [11] showed that Strassen's characteristic add/subtract steps can be fused directly into an already-optimized DGEMM's packing routines, needing no workspace beyond what packing already allocates, and that this makes Strassen practical at far smaller N (and for skinnier rank-k updates) than prior conventional wisdom held; a GPU follow-up extends the same idea to CUDA [12]. Ballard, Demmel, Holtz & Schwartz [7] give the communication-optimal theoretical counterpart, proving inter-processor communication lower bounds for Strassen-based parallel algorithms and giving an algorithm that matches them. This project contributes two independent Strassen rebuilds sharing one measurement harness – a classical rebuild (Section 7) and the 2024 alternate-basis variant (Section 6) – offered as corroborating, independently- built implementations rather than as claiming to outperform these results.

2.3 Numerical accuracy of fast matrix multiplication

It is established theory that the classical triple-loop algorithm satisfies a component-wise error bound while Strassen and other fast algorithms satisfy only a weaker norm-wise bound – a gap our own earlier accuracy sweep (FINDINGS.md Part I) demonstrates concretely, measuring a >1400% *max-componentwise*-relative error for a 2-level classical-Strassen rebuild under row-scaled ill-conditioning at N=1024, while the Frobenius-relative error for the same run stays near 10^{-15} – a live illustration of exactly why the two metrics can disagree. Dumas, Pernet & Sedoglavic [9] (2024) give a 7-multiplication alternate-basis algorithm with a provably lower error-growth factor ($\gamma_{2,1} \approx 12.066$ versus classical Strassen's ≈ 14.828) and report better accuracy in practice on random test matrices. Section 6 is, to our knowledge, its first hardware-measured benchmark under realistic packed/blocked, multi-threaded execution.

2.4 Measurement methodology and reproducibility

Mytkowicz, Diwan, Hauswirth & Sweeney [13] showed that "insignificant" experimental-setup choices (environment variable size, link order) can flip a systems benchmark's conclusion, and gave both a detection method (causal analysis) and a mitigation (setup randomization). Section 5 is a domain-specific instance of exactly this phenomenon, applied for the first time (as far as we found) to a DGEMM/OpenMP scheduling claim. The Roofline model [15] is the standard compute/bandwidth ceiling framework our own compute-ceiling and STREAM-bandwidth analysis is

built on. Davis (2025) [8] applies bootstrap 95% confidence intervals and a Games-Howell pairwise-significance test to GEMM benchmarking across 30 trials per configuration – the statistical standard Section 3 adopts as the target protocol, not yet retrofitted onto every headline number in this draft (see Section 8). We are careful to distinguish our own sense of "reproducible" (a conclusion that survives repeated, controlled measurement) from two adjacent but different senses already in the literature: GEMMbench [2] is a crowdsourced benchmark-*collection* framework, and a separate line of work on "reproducible" GEMM concerns bit-for-bit reproducibility despite floating-point non-associativity and dynamic scheduling – neither is the measurement-validity question this paper asks, and the three should not be conflated. The broader reproducibility/artifact-evaluation trend in systems and HPC research [5, 3, 1] – SC's mandatory Artifact Description appendix since 2019, ACM TOMS's Replicated Computational Results designation – is the concrete incentive structure Section 9's artifact targets, and evidence that artifact availability correlates with citation and reuse is part of why we frame this paper around a public, reproducible harness rather than a fixed set of numbers.

2.5 LLM-driven kernel generation (forward-looking)

A fast-growing 2025–2026 literature generates GEMM and other kernels directly via large language models [4, 6], typically evaluated on single or best-of-N wall-clock numbers rather than the statistical/hardware-counter rigor this paper's harness provides throughout. We do not build this out as a contribution here – see Section 10 for why it is left as one paragraph of future work rather than a section of its own.

3 Experimental Methodology

3.1 Environments

No machine used anywhere in this project is the original paper's target platform (a dual-socket Intel Xeon Gold 5218 / Cascade Lake cluster node with AVX-512), and AVX-512 hardware was not reachable at any point in this project – stated plainly here and returned to in Section 8, not glossed over. Two machines were used throughout:

- **Development machine**: a single-socket Intel Core i7-13700K (8 performance cores + 8 efficiency cores, 24 logical processors via hyperthreading on the P-cores only, 64 GB RAM, single NUMA node), accessed three ways – natively under Windows (MSVC 19.44, no virtualization layer, the closest available substitute for a genuinely bare-metal environment), under WSL2/Hyper-V (gcc), and under WSL2 + Docker. `cpuinfo` confirms only `avx`/`avx2`/`fma` flags – Intel fuses off AVX-512 on this consumer part once efficiency cores are active. Hardware performance counters (`perf stat`) are unavailable under WSL2/Hyper-V entirely, confirmed independently via direct testing and via a toolchain sanity check, with every event reporting `<not supported>` even inside a `--privileged` Docker container. WSL2/Hyper-V also does not expose a stable guest-vCPU-to-physical-core mapping: a pure-FMA microbenchmark pinned to each of the 24 logical CPUs via `taskset` inside WSL2 measured a flat ≈70–71 GFLOP/s everywhere (no bimodal P/E split); the identical test run natively via `SetThreadAffinityMask` showed a clean split (logical CPUs 0–15, the P-core/hyperthread pairs, ≈33–34 GFLOP/s; logical CPUs 16–23, the E-cores, ≈17 GFLOP/s) – so any core-placement experiment in this paper that depends on P/E identity necessarily runs natively, not under WSL2/Hyper-V.

- **`glogin.cs.vt.edu`**: a KVM-virtualized 8-vCPU slice (2 vSockets × 4 cores, no hyperthreading) of an Intel Xeon E5-2630 v3 (Haswell-EP, 2015; `avx`/`avx2`/ `fma` only), no sudo access, gcc 11.5.0 (matching the audited paper's own compiler version), with Clang 22.1.3 and a complete Intel oneAPI 2022.1 install (`icx`, MKL, VTune, `llvm-mca`) also available. Real hardware PMU counters are obtainable here, unlike under WSL2/Hyper-V, though this PMU does not expose LLC-specific events (unlike the audited paper's Cascade Lake target), so generic `cache-references`/ `cache-misses` and `L1-dcache-*` events substitute throughout Section 7's root-cause analysis.

3.2 The repeated-measurement protocol

Section 5 is itself the motivating demonstration for this subsection, not just an application of a rule stated in advance: an initial single-run comparison at N=1024, T=8 measured *unpinned* `static` scheduling at 218.0 GFLOP/s against *unpinned* `dynamic` at 104.4 GFLOP/s – a clean, apparently real 2× effect. Four repeats of each configuration measured `static` $\in \{277, 287, 217, 296\}$ and `dynamic` $\in \{285, 276, 294, 286\}$ GFLOP/s – fully overlapping ranges with no consistent winner; the original "218 vs. 104" comparison was simply two single draws landing on opposite tails of one wide (≈217–296) distribution. Every scheduling- or placement-related claim in this paper is held to that standard from this point on: $n \geq 4$–5 independent repeats minimum, reporting the full range, before any comparison is stated as a finding rather than a single-run observation. The target protocol going forward, following Davis (2025) [8], is **[TODO: settle on $n = 20$–30]** repetitions per configuration with bootstrap 95% confidence intervals and a Games-Howell (or equivalent) pairwise-significance test; as of this draft that protocol is adopted for Section 5's headline comparisons (Section 5.4) but not yet retrofitted onto every supporting-material number in Section 7 – see Section 8.

3.3 Instrumentation

Three complementary instrumentation layers are used throughout: (1) opt-in call/byte counters (`harness/count_calls.h`, enabled via `-DCOUNT_CALLS`, confirmed to add no measurable timing overhead or behavioral change when disabled) for recursive-call and allocation accounting; (2) `perf stat` hardware counters where available (cycles, instructions, branches/branch-misses, cache-references/misses, L1-dcache loads/misses, dTLB loads/misses), supplemented by Intel VTune Profiler's driverless "User-mode sampling and tracing" mode on `glogin`; (3) phase-separated software timers (`avx2/ladder/timer.h`, cross-platform via `QueryPerformanceCounter`/`clock_gettime`) splitting execution into allocation, A-panel packing, B-panel packing, kernel compute, kernel write-back, and (for the Strassen variants) add/subtract, combine, workspace bookkeeping, and task-scheduling phases. Several hard environment limits were confirmed directly rather than assumed: VTune's `memory-access` collection mode refuses outright inside `glogin`'s VM (*"Memory Access analysis is not supported inside a virtual machine since uncore events cannot be collected"*); `uarch-exploration` needs `perf_event_paranoid` ≤ 0 or a kernel sampling driver, both requiring root, unavailable; and `/sys/class/powercap/intel-rapl*` does not exist on that VM at all, so no RAPL power telemetry was obtainable anywhere in this project (Section 8).

4 Correctness Audit

We audited the paper's own codebase (`avx/`, `Parallel/`, `test/`) against its own description, using two independent verification paths since no AVX-512 hardware was available: a temporary scalar stand-in kernel (same packed-buffer contract, no intrinsics, not kept in the repository afterward)

that let the real packing/blocking control logic execute and be checked on any CPU during the audit itself, and a permanent AVX2 port (`avx/kernel_avx2.c`, `avx/strassen_avx2.c`, confirmed via `objdump` to emit genuine `vfmadd*pd` instructions) that gives durable, re-runnable, genuinely vectorized correctness coverage going forward. Every fix below was verified by real execution, not by reading the code and reasoning about it.

Table 1: Correctness bugs found and fixed in the audited codebase. Full evidence (command output, backtraces, before/after) in `revalidation/correctness/2026-09-09_bug_evidence_and_fixes.md`.

Bug	File(s)	Root cause	Fix
Computed $C{=}B{\cdot}A$, not $A{\cdot}B$	`avx/matmul.c`	Packing routines read A/B with contraction and output row/column roles swapped; the existing test only passed because its own test matrices are symmetric (a false-positive test)	Rewrote packing with explicit row-major offsets; added an $O(N^2)$ transpose pass
Data race in the blocked parallel kernel	`Parallel/mm_parallel_blocked*.c`	`collapse(3)` parallelized over the reduction dimension with unsynchronized `+=`; only "worked" because one compiler's static scheduling happened to keep racy iterations non-overlapping	`collapse(3)` → `collapse(2)`; verified correct under adversarial `schedule(dynamic,1)`
Silent data loss for non-power-of-2 $N{>}128$	`avx/strassen.c`	Recursion halves N with no remainder handling, unlike every other Strassen variant in the repo, which all pad first	Pad to the next power of two before recursing
Segfault on a clean checkout	`test/main.c`, `test/run.sh`	Created one output directory but not two others the code writes to; every driver function `fprintf`s to an unchecked `fopen()` result	Added the missing directory-creation calls

The most consequential of the four is the first: a packing bug that computes the transpose of the intended product for non-square/ non-tile-aligned shapes, which no test in the original repository could have caught because its own correctness test happens to use symmetric matrices. **[TODO: Insert a short worked numerical example (e.g., a small non-symmetric A, B where $C_{\text{buggy}} = BA \neq AB = C_{\text{correct}}$) for concreteness.]**

A second, repo-wide review pass at higher scrutiny found and fixed six further defects beyond the four above, three worth calling out specifically: a heap overflow for $N \leq 0$ and small powers of two in the Strassen padding logic added to fix bug 3 in Table 1 – *a regression introduced by this audit's own first-pass fix*, only surfaced by a second, more thorough pass, and worth noting as a general lesson that a fix itself needs re-review, not just the code it touched; an analogous infinite-recursion-at-$n{=}1$ bug in two other Strassen variants sharing the same unfloored padding logic; and four unchecked Win32 API calls in the native core-topology classifier (`SetThreadAffinityMask`, `CreateThread`, `WaitForMultipleObjects`, and a bit-shift that overflowed for CPU index $\geq$64) whose silent failure modes would have produced plausible-looking but wrong core-utilization numbers – directly relevant to Section 5's core-placement measurements, which depend on that same classifier being correct. Four further issues were found and deliberately left unfixed, each with a stated reason (memory-cost/complexity trade-offs, an intentional drop-in-alternative symbol design, and one ambiguous-intent global-state question) rather than silently dropped or guessed at – see

revalidation/correctness/2026-09-09_bug_evidence_and_fixes.md for the full reasoning.

This is, deliberately, exactly the shape of contribution ACM TOMS's Replicated Computational Results designation and SC's Artifact Description requirement were built to reward (Section 2): an independent re-execution of a specific published implementation, with real, reproducible evidence of divergence from its own claims.

5 Is the Scheduling Win Real, or Virtualization Compensation?

5.1 The claim under test

An earlier stage of this project's own AVX2 DGEMM work measured a large OpenMP `schedule(dynamic)` win over `schedule(static)` under WSL2/Hyper-V, reported at the time with an explicit caveat that it would need confirmation on non-virtualized hardware before being generalized. This section is that confirmation.

5.2 Method

A single portable C source (`avx2/ladder/matmul_ladder.c`, MR=4×NR=8 broadcast-A/vector-B kernel) compiles unmodified under both MSVC 19.44 (OpenMP 2.0, no `collapse`, no runtime schedule-selection API, C99 loop-variable declarations rejected inside `#pragma omp for`) and gcc, selecting the OpenMP schedule at *runtime* via `schedule(runtime)` plus the `OMP_SCHEDULE` environment variable – portable across both compilers' OpenMP feature levels with no recompilation needed to compare scheduling policies. Phase timers instrument packing and kernel-compute time separately (`pack_a_s`, `pack_b_s`, `kernel_s`) around a real multilevel blocking driver, and (Windows only) explicit per-thread `SetThreadAffinityMask` pinning to a caller-supplied logical-CPU list enables the controlled core-placement experiment below.

The harness's first build itself produced one instructive methodology bug: a kernel-only stage whose output tile was never read reported an impossible 21,845,696 GFLOP/s under WSL2/gcc, because `-O3` proved the repeated-call loop had no observable effect and deleted it entirely (MSVC did not perform this optimization – itself a real, if incidental, cross-compiler data point). Fixed with a standard `volatile` anti-dead-code-elimination sink; we record it here precisely because it is exactly the class of silent, plausible-looking methodology bug this section's broader argument is about.

5.3 An initial cross-environment sweep, and why it is not trustworthy on its own

A single-run sweep of $N \in \{512, 1024, 2048\}$, $T \in \{1, 4, 8\}$, schedule $\in$ {`static`, `dynamic`(chunk=4), `guided`} across native Windows, WSL2/Hyper-V, and `glogin` (KVM) shows a clear qualitative split: on `glogin`, scheduling choice barely moves the needle at any thread count (e.g. N=2048, T=8: `static` 118.4, `dynamic` 119.4, `guided` 119.4 GFLOP/s – within 1%, at every (N, T) tested), consistent with a stable, non-reshuffling vCPU-to-pCPU mapping under KVM. On WSL2 and native Windows, the single-run sweep shows apparent scheduling differences with *no stable direction* – dynamic wins at one (N, T), static wins at the next, for the same schedule pair on the same machine. Read uncritically, this pattern would seem to confirm the original dynamic-scheduling finding; Section 3.2's protocol is precisely why we do not read it that way without repeating it first. We also flag, rather than quietly discard, an unexplained discrepancy in this sweep: the same N=1024, T=8 Windows configuration measured

roughly 2× lower in this back-to-back 27-configuration sweep than in the isolated repeated-run experiment below, run minutes apart on the same binary and machine – the leading hypothesis is thermal or sustained-load throttling accumulating over the sweep, but this was not confirmed (no frequency/temperature telemetry was collected in this pass). Every number from this sweep is therefore treated as lower-confidence than the controlled experiment that follows.

5.4 The controlled experiment: core placement, properly repeated

Native Windows is the only environment in this project offering working thread-to-core pinning, so it is the only one in which the actual "is it scheduling policy, or is it core heterogeneity" question can be asked cleanly, using the P-core/E-core classification from Section 3.1 (logical CPUs 0–15 = P-cores/hyperthread pairs, 16–23 = E-cores). At N=1024, T=8, a first single-run pass appeared to show two striking effects: unpinned `static` (218.0 GFLOP/s) beating unpinned `dynamic` (104.4 GFLOP/s) by roughly 2×, and `dynamic` edging out `static` on a mixed 4 P-core + 4 E-core placement – exactly the heterogeneous-core scenario scheduling theory predicts dynamic scheduling should help most.

Neither survived repetition. Four repeats of the unpinned comparison gave `static` $\in \{277, 287, 217, 296\}$ and `dynamic` $\in \{285, 276, 294, 286\}$ GFLOP/s – completely overlapping, no consistent winner; the original 218-vs-104 result was noise from two single draws landing on opposite tails of one wide distribution. Five repeats of the mixed-placement comparison gave `static` $\in \{150, 141, 143, 151, 149\}$ and `dynamic` $\in \{146, 149, 145, 151, 145\}$ GFLOP/s – again completely overlapping, even in the one scenario chosen specifically because theory favors dynamic scheduling there. Notably, this pinned distribution (141–151) is far tighter than the unpinned one (217–296): pinning removed most of the variance, but did not reveal a scheduling-policy effect underneath it – it simply removed OS-placement noise.

What *did* produce a large, clean, low-variance effect across every repeat: which physical cores the threads land on at all. Pinned-to- P-cores (≈250–280 GFLOP/s) beats pinned-to-E-cores (≈133–137 GFLOP/s) by roughly 2×, consistently, regardless of scheduling policy – mechanistically unsurprising given P-cores' roughly 2× per-cycle AVX2 throughput advantage (Section 3.1), and it dwarfs any scheduling-policy effect measured anywhere in this experiment.

5.5 Conclusion

On this hardware, in this pass, properly repeated measurement finds no detectable OpenMP scheduling-policy effect once thread-to-core placement is controlled for – including in the heterogeneous-core case scheduling theory says should favor dynamic scheduling the most. This does not prove the original WSL2 result was purely a virtualization artifact (that would require the same repeated-measurement rigor applied *inside* WSL2/Hyper-V itself, which Table 1's sibling open-items list in Section 8 still flags as undone); it does mean the evidence collected so far does not meet the burden of proof for treating dynamic scheduling as architecturally necessary, and the one clean non-virtualized test available found no such effect. We recommend, and adopt in our own reporting going forward, scoping any such scheduling-driven performance figure explicitly to the environment it was measured under, and treating figures obtained purely from architecture-level, packing/blocking work (not scheduling policy) as the portable, generalizable claim.

6 The Dumas et al. (2024) Alternate-Basis Strassen, Implemented and Measured

6.1 Getting the formula right

Dumas, Pernet & Sedoglavic [9] state their accuracy-optimized 2×2 algorithm (their Equation 36, instantiated concretely in their Table 3) only as a straight-line program, in the paper's own a_{ij}/b_{ij} subscript convention. A literal transcription against standard row-major indexing produces an *incorrect* result – off by exactly the antisymmetric $a_{12} \leftrightarrow a_{21}$, $b_{12} \leftrightarrow b_{21}$ terms. We caught this symbolically (`sympy`; `revalidation/dumas2024/verify_formula.py`) *before* writing any implementation code, by expanding the literal transcription against the reference product $C = AB$, finding four nonzero symbolic residuals, and brute-forcing input/output relabelings until the one that zeroes all four was found. The corrected formula (block-generalized; every a_{ij}/b_{ij} below is a $k \times k$ submatrix) is:

$$
\begin{aligned}
t_1 &= \tfrac{\sqrt{3}}{3} A_{22}, & s_1 &= \tfrac{\sqrt{3}}{3} B_{12}, \\
t_2 &= A_{12} + t_1, & s_2 &= s_1 - B_{11}, \\
t_3 &= A_{21} + t_2, & s_3 &= s_2 + B_{22},
\end{aligned}
$$

$$
\begin{aligned}
\ell_1 &= \tfrac{\sqrt{3}}{2} A_{11} + \tfrac{1}{2} t_3, & r_1 &= 2 s_1, \\
\ell_2 &= A_{21} - t_1, & r_2 &= s_2, \\
\ell_3 &= t_2, & r_3 &= s_1 - B_{22}, \\
\ell_4 &= 2 t_1, & r_4 &= \tfrac{1}{2} s_3 - \tfrac{\sqrt{3}}{2} B_{21}, \\
\ell_5 &= \ell_2 - \ell_1, & r_5 &= r_3 + r_4, \\
\ell_6 &= \ell_5 + \ell_4, & r_6 &= r_1 - r_5, \\
\ell_7 &= \ell_5 + \ell_3, & r_7 &= r_5 - r_2,
\end{aligned}
$$

$$
p_i = \ell_i \cdot r_i \quad (i = 1, \ldots, 7),
$$

$$
\begin{aligned}
w_2 &= p_5 + p_1 + p_6, \qquad & w_1 &= p_7 + p_6, \qquad & w_3 &= w_2 - p_2, \qquad & w_5 &= \tfrac{1}{2}(p_4 + w_2), \\
C_{12} &= p_1 - p_3 - w_5, \qquad & C_{21} &= w_3 - w_5, \qquad & C_{22} &= \sqrt{3}\, w_5, \qquad & C_{11} &= \tfrac{\sqrt{3}}{3}(w_3 - C_{12} - 2 w_1).
\end{aligned}
$$

The algorithm is non-commutative by construction (its authors' own characterization), which is precisely the property that makes this scalar 7-multiplication formula safe to reuse recursively at the block level, the same way classical Strassen's is; Section 6.3 confirms this empirically rather than assuming it from the paper's claim alone.

6.2 Implementation

Implemented in `avx2/ladder/dumas_strassen_rebuild.c` on infrastructure identical to the classical-Strassen rebuild used as baseline throughout this section (`strassen_rebuild.c`): the same bump-pointer pooled arena, the same per-level disjoint sub-arena carving for concurrent `#pragma omp task` siblings (avoiding an aliasing hazard a naively shared arena copy would introduce under real task parallelism), and the same phase-timing instrumentation (leaf/add-subtract/combine/workspace/scheduling). *The only variable between the two binaries is the 7-multiplication formula itself* – this isolation is what makes Section 6.5's performance comparison attributable specifically to the formula rather than to incidental implementation differences.

6.3 Correctness

Run at $N \in \{64, 128, 256, 512\}$, recursion depth 1–3, against an independent long-double-accumulation reference: relative error 4.7×10^{-16} to 1.8×10^{-15} at every configuration – machine precision. This is, to our knowledge, the first execution-based confirmation that the 2024 algorithm's block/recursive generalization is correct, not only its scalar 2×2 base case that the original paper analyzes.

6.4 Accuracy

Extending the accuracy harness used for classical Strassen (five input distributions: uniform, gaussian, wide-dynamic-range, ill-conditioned, cancellation-heavy) to also run the Dumas variant at depths 1–3, N=512:

Table 2: Frobenius-relative error, classical Strassen vs. the Dumas et al. (2024) variant, depth 3, N=512 (`revalidation/dumas2024/accuracy_results_N512.csv`; depths 1–2 show the same qualitative pattern, omitted here for space).

Distribution	Classical Strassen	Dumas (2024)
uniform	2.20×10^{-15}	1.39×10^{-15}
gaussian	2.27×10^{-15}	1.40×10^{-15}
wide_range	2.30×10^{-15}	1.39×10^{-15}
ill_cond	4.05×10^{-15}	2.45×10^{-15}
cancel	1.08×10^{-15}	4.39×10^{-16}

Frobenius-relative error is consistently *lower* for the Dumas variant than for classical Strassen at every matched recursion depth, across all five distributions – e.g. ≈38% lower under the gaussian distribution at depth 3. This qualitatively confirms [9]'s claim, now under real packing/blocking/task parallelism the original paper's own evaluation did not use. Max-componentwise-relative error (not tabulated here; see the source CSV) is a noisier, mixed picture – consistent with the paper's own bound being a Frobenius/2-norm-type bound ($\gamma_{2,1}$), not a componentwise one, so the advantage should be expected to appear cleanest exactly where it does. At these matrix sizes and this uncorrelated-random distribution design, neither algorithm approaches the >1400% catastrophic error Section 7 measures under *row-scaled, correlated* ill-conditioning specifically – whether the Dumas variant's improved accuracy narrows that particular catastrophic case remains open (Section 10).

6.5 Performance: the accuracy improvement is not free

Head-to-head phase-timed comparison against classical Strassen, identical leaf kernel/blocking parameters/thread count, at $N \in \{512, 1024, 2048\} \times \text{depth} \in \{1, 2, 3\}$ (`revalidation/dumas2024/perf_comparison_results.csv`):

`leaf_s` matches within run-to-run noise at every depth (confirming the control); `addsub_s` is consistently and substantially higher for the Dumas variant (≈57% higher at depth 3), driven by its 24-operation straight-line program (several irrational-$\sqrt{3}$-coefficient scale-and-combine steps) against classical Strassen's ≈18 plain adds/subtracts. Net effect: a ≈20.7% GFLOP/s cost at N=2048, depth 3 for the accuracy improvement measured above; the cost shrinks at shallower depth (≈6.4% at depth 1, fewer combine levels executed). As far as we found, *neither the original paper (which reports only asymptotic operation counts and an accuracy-vs-size plot) nor any other publication reports this accuracy/performance trade-off.*

Table 3: Classical Strassen vs. the Dumas et al. (2024) variant at $N = 2048$, $T = 8$ (dev machine, 5-repetition best-of; not yet bootstrap- CI'd, see Section 8). `leaf_s` is the shared, unmodified leaf kernel – the control confirming the slowdown is attributable to the combine/scale formula, not the harness.

Depth	Algorithm	leaf_s (s)	addsub_s (s)	GFLOP/s
1	Classical Strassen	0.288	0.041	50.79
1	Dumas (2024)	0.295	0.058	47.56
2	Classical Strassen	0.268	0.109	42.76
2	Dumas (2024)	0.273	0.155	38.16
3	Classical Strassen	0.293	0.259	127.67
3	Dumas (2024)	0.301	0.408	101.27

7 Supporting Material

7.1 A from-scratch AVX2 DGEMM vs. production BLAS

Independent of the audited codebase, we built a new AVX2 DGEMM from scratch and tuned it via a disassembly-driven ablation series (operand-swap ordering, unroll-plus-prefetch, microkernel shape selection, hand-written assembly, vectorized operand packing), reaching ≈87–92% of production OpenBLAS/BLIS/MKL throughput – the figure we treat as the architecture-level, portable claim. A dynamic- scheduling variant on top measured 117–130% of production BLAS under WSL2/Hyper-V; per Section 5's finding, we now scope that figure explicitly to "measured under WSL2/Hyper-V" rather than generalizing it, since the scheduling-policy mechanism behind it did not survive repeated, controlled measurement elsewhere in this project.

7.2 Hardware-counter root-cause analysis

Using hardware performance counters on `glogin` (unavailable under WSL2/Hyper-V; Section 3.1) to explain, not just report, several classical DGEMM performance results: naive triple-loop DGEMM's well-known performance cliff at scale is driven by IPC cratering from ≈1.3 to ≈0.65 as N crosses 256, with branch-miss rate *falling* and LLC-level miss rate staying near zero throughout – ruling out both branch misprediction and DRAM-capacity misses, and pinning the cause specifically to repeated L2/L3-latency exposure from B's column-access pattern, a blocking/reuse-distance problem rather than a software-prefetch-for-DRAM-latency one. Blocking's own advantage is not explained by a lower L1-miss rate in isolation (blocked's 48% L1-miss rate is barely better than naive's 63%) but by *where* those misses resolve: blocking keeps the active working set cycling through cache fast enough that most L1 misses hit nearby in L2 rather than paying full round-trip latency, yielding a 3.6× IPC improvement despite the similar raw miss rate. Pure (uncut-off) Strassen has the *best* IPC and cache behavior of every algorithm measured, and its cost is purely instruction/allocation volume: 19,608 recursive calls at N=64 growing to 6,725,601 at N=512, exactly matching the audited paper's own reported figures, directly confirming the paper's own qualitative diagnosis ("excessive recursion," "allocator pressure") with hard counter evidence rather than just call-count bookkeeping, and explaining why parallelizing it makes things *worse* (more threads means more concurrent allocator contention layered on an already overhead-dominated workload). At N=1024, a loop-reordered-and-unrolled hybrid-Strassen variant beats a blocked-and-unrolled classical variant on every axis simultaneously (0.693s vs. 2.329s wall-clock, 1.69 vs. 1.28 IPC, 25.0% vs. 40.6% L1-miss rate) – a genuine compounding advantage, and the hard-counter confirmation

behind our own cross-machine finding (Table 4) that the audited paper's scalar-level "blocked wins for n>512" claim does not generalize.

Table 4: Blocked-and-unrolled vs. hybrid-Strassen-unrolled, scalar (non-SIMD), local i7-13700K, serial, best-of-3. The audited paper's own crossover claim does not hold at this (scalar) comparison level on two independent non-target machines; it does hold once both sides are AVX-vectorized, at a smaller margin (see main text).

N	blocked_unrolled (s)	strassen_base_unrolled (s)	Strassen's margin
512	0.059	0.028	2.1×
1024	0.411	0.141	2.9×
2048	3.262	0.812	4.0×

7.3 Cross-compiler codegen

Building the identical portable harness with gcc 11.5.0, Clang 22.1.3, and Intel `icx` 2022.1.0 (all available on `glogin`, same optimization flags, 5 repeats each) shows Clang winning the full N=1024, T=8, `static` pipeline by ≈20% (mean 140.2 vs. gcc's 116.9 GFLOP/s) despite gcc having a marginally *higher* isolated kernel-only number (mean 28.8 vs. 27.8 GFLOP/s) – meaning the compiler difference lives outside the hot FMA loop, not inside it. Disassembly (with the kernel forced `noinline` so the symbol survives comparably across all three compilers, since Clang/icx fully inline it by default and gcc does not) confirms one real, mechanistic difference: Clang and icx both auto-unroll the k-loop 2× at `-O3`; gcc does not, the same transformation our own AVX2 ablation series had to add by hand for gcc. Checked quantitatively with `llvm-mca -mcpu=haswell` on the extracted loop bodies, per-k-step modeled throughput is nearly identical (gcc: 5.14 cycles/step; Clang unrolled: 5.08 cycles/step) – so the confirmed auto-unrolling is *not* the primary explanation for the measured 20% full-pipeline gap, which is specifically a multi-threaded effect (gcc is actually faster at T=1, with higher IPC: 2.41 vs. Clang's 2.01) whose full mechanism (most likely candidate: per-thread load- imbalance or false-sharing differences in packing-loop dispatch) was not fully pinned down in this pass.

7.4 Classical-Strassen rebuild: a measured optimized-vs-optimized crossover

A from-scratch Strassen rebuild (pooled single-allocation workspace per recursion level, only the 10 genuinely-combined operands ever materialized, the 4 quadrants used bare passed as strided views with no copy, thread-aware task-vs-sequential cutoff) reports throughput using the classical $2N^3$ FLOP formula for direct comparability, deliberately labeled "effective" to avoid overstating Strassen's true (lower) FLOP count:

A 2-level variant is consistently worse at every size tested (recursion overhead compounds faster than arithmetic savings materialize) and never reaches a crossover in this range. A companion accuracy sweep on this same rebuild is what produces this paper's headline numerical- accuracy finding: under row-scaled ill-conditioning, 1-level Strassen's max-componentwise-relative error reaches 84% at N=1024 while 2-level reaches **1436%** (i.e. worse than the largest element being completely wrong), while classical DGEMM stays at $\approx 10^{-10}$ throughout the same sweep – confirming, with a dramatic and reproducible number, exactly the accuracy concern the audited paper cites as a reason to avoid fast-multiplication approaches without ever measuring it for its own implementation.

Table 5: Optimized classical DGEMM vs. optimized 1-level Strassen, effective GFLOP/s (= $2N^3$/time). A genuine, measured crossover at N=8192, not the "never crosses" story an unoptimized-vs- optimized comparison would tell.

N	DGEMM baseline	Strassen (1-level)	Ratio
512	245.8	22.9	10.8×
1024	466.9	105.6	4.4×
2048	458.2	260.0	1.76×
4096	434.7	375.0	1.16×
8192	426.6	426.7	≈1.00× (crossover)

7.5 Rectangular and irregular shapes

Generalizing the portable harness from square-only to arbitrary $M \times K \times N$ (verified correct, 0.000e+00 relative error against an independent long-double reference, across tall-skinny, short-wide, large-K, and deliberately awkward sizes) surfaces a real performance cliff, not just a correctness edge case: at N=1024 (an exact multiple of the microkernel's 4×8 tile shape), throughput is ≈111–125 GFLOP/s depending on blocking parameters; at N=1023, throughput drops to **56.7 GFLOP/s** – roughly half – and at N=1025, to **50.8 GFLOP/s**, worse still. The mechanism is direct: every tile touching a boundary needs the masked/remainder write-back path instead of the fast full-tile path, and for a size one off a clean multiple, most tiles along that edge hit it. This is a first-order consideration for choosing microkernel tile shapes (or a remainder strategy) in any production context where inputs are not guaranteed to land on a tile boundary, and is not something a square-power-of-two-only benchmark would ever surface.

8 Discussion and Threats to Validity

We hold ourselves to the same standard of stating limitations plainly that motivates Section 5: no AVX-512 hardware was reachable at any point in this project, so every correctness result in Section 4 rests on a temporary scalar stand-in kernel plus a permanent, execution-verified AVX2 port, not on AVX-512 execution itself – a real limitation, not one we found a way around. No literal bare-metal Linux machine was available; native Windows (no virtualization layer) is the closest non-Hyper-V substitute reached, and Section 5's controlled experiment is confined to that platform because it is the only one offering working thread-to-core pinning in this project – the specific claim that a WSL2/Hyper-V-measured scheduling effect is a pure virtualization artifact (rather than merely unreproduced elsewhere) would require the same repeated-measurement rigor applied *inside* WSL2/Hyper-V itself, not yet done. RAPL power and CPU-frequency telemetry were unavailable on every VM tested (confirmed via a direct `/sys/class/powercap` check and, independently, via VTune's own refusal to run certain collection modes inside a VM – Section 3) and not yet attempted on native Windows. The single-socket, single-NUMA-node hardware available throughout this project means none of the audited paper's dual-socket NUMA claims could be tested at all, not merely under a configuration constraint. Section 3.2's bootstrap-CI/significance-test protocol has been adopted for Section 5's headline comparisons but not yet retrofitted onto every number in Section 7 – those figures should be read as directionally trustworthy (several are corroborated by independent hardware-counter evidence explaining the mechanism, not just a wall-clock number) but not yet held to the same statistical standard. Finally, Section 6's performance comparison (Table 3) has so far been run only on the development machine via WSL2; cross-machine

confirmation on `glogin` and native Windows, using the same harness already proven portable in Section 5, is listed as future work (Section 10) rather than claimed here.

9 Reproducibility and Artifact Availability

Every number in this paper traces back to a named, re-runnable script and a checked-in CSV or log in this project's repository, organized so results from different sessions and hardware can be compared later without re-deriving them or trusting memory: `revalidation/correctness/` holds the Section 4 bug evidence; `revalidation/benchmarks/` and `revalidation/harness/` hold the cross-machine wall-clock and hardware-counter data behind Section 7; `avx2/ladder/` holds the Section 5 measurement- bias harness and its raw sweep/pinning-experiment CSVs; and `revalidation/dumas2024/` holds the Section 6 formula-verification script, correctness sweep, and accuracy/performance comparison data. Each directory's own `README.md` documents exact machine/compiler/command-line provenance for its data. **[TODO: Remaining step, per FINDINGS.md Part L.3.2: write a single top-level reproducibility README tying these together end-to-end, add a license file, and cut a versioned, DOI'd release (e.g. via Zenodo) so the artifact is independently citable regardless of where this paper's prose is eventually published.]** Target venue is not yet finalized; FINDINGS.md Part L.3 recommends, in order of fit: ACM TOMS's Replicated Computational Results track (closest fit for Section 4 alone), the PMBS workshop (SC-colocated; closest fit for Section 5), or an arXiv preprint immediately regardless of either.

10 Conclusion and Future Work

We set out to check whether a published AVX-512 DGEMM/Strassen implementation does what it claims, whether its reported performance wins are real hardware effects, and whether a very recent (2024) result in the numerical-accuracy literature could be benchmarked for the first time using the same infrastructure. On the first question, we found and fixed four real correctness bugs (plus six more in a deeper second pass), the most serious silently transposing the product of a non-square multiplication. On the second, a large scheduling-policy performance win did not survive proper repeated measurement, while raw core-placement effects did – a concrete, domain-specific demonstration that measurement bias (Mytkowicz et al., 2009) remains a live concern in exactly the kind of virtualized development environment most GEMM tuning work is now done in. On the third, we implemented and measured Dumas et al.'s 2024 alternate-basis Strassen variant for what appears to be the first time under realistic execution, confirming its accuracy claim and quantifying, for the first time, the performance cost that comes with it.

Concrete next steps, in priority order: cross-machine confirmation of Table 3's performance-cost figure on `glogin` and native Windows; applying Section 7's row- correlated catastrophic-ill-conditioning construction to the Dumas variant, to determine whether its improved accuracy meaningfully narrows that specific catastrophic case or is orthogonal to it; project- wide adoption of Section 3.2's bootstrap-CI protocol beyond Section 5; and running the measurement-bias protocol from Section 5 *inside* WSL2/Hyper-V itself (not just outside it), to settle directly whether the original scheduling win was virtualization-compensation rather than merely unreproduced elsewhere. One further direction, noted but deliberately not built out here: the same statistical/hardware-counter measurement rigor this paper applies throughout is largely absent from the fast- growing LLM-driven kernel-generation literature (Section 2), which is presently evaluated mostly on single or best-of-N wall-clock numbers; using this project's harness to evaluate an LLM-generated kernel

under the same protocol would be a natural, low-risk extension, left for future work rather than claimed as a contribution of this paper.

Acknowledgments

[TODO:]

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